

Leaky Integrate-and-Fire (LIF)

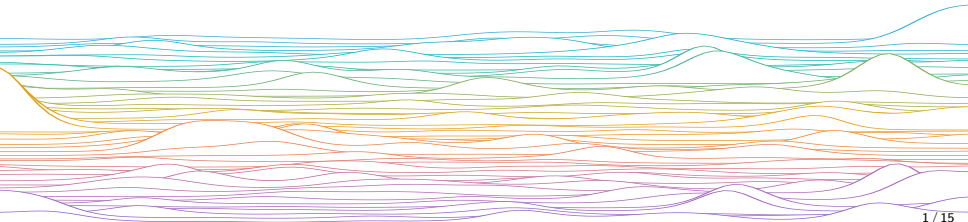
Part I: Overview

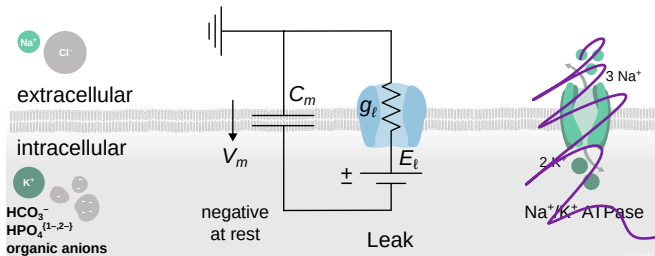
Computational Neuroscience
University of Bristol

M Rule

Learning outcomes:

- ▶ Mathematically analyze spiking dynamics of the LIF model
- ▶ The f - I curve (rate model) for LIF





$$C_m \dot{v} = g_\ell (E_\ell - v)$$

$$\tau_m \dot{v} = E_\ell - v$$

$$C_m = 100 \times 10^{-12} \text{ Farads} \quad (100 \text{ picofarads})$$

cell membrane capacitance

$$E_\ell = -70 \times 10^{-3} \text{ Volts} \quad (-70 \text{ millivolts})$$

resting potential

$$g_\ell = 10 \times 10^{-9} \text{ Siemens} \quad (10 \text{ nanosiemens})$$

leak conductance

$$R_m = 1/g_\ell = 100 \times 10^6 \text{ Ohms} \quad (100 \text{ megaohms})$$

membrane resistance

$$\tau_m = r_m C_m = 10 \times 10^{-3} \text{ seconds} \quad (10 \text{ milliseconds})$$

membrane time constant

Modelling Scales

... Quantum Chemistry, Molecular dynamics

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Molecules
Gillespie, Master Equation
Concentrations
Mass-Action Kinetics
Conductance Models
Hodgkin–Huxley
Spiking Models
Leaky Integrate and Fire
Rate Neurons
Neural Mass/Field Models
Poisson Neurons
Generalized Linear Models
Binary Neurons
McCulloch–Pitts, Hopfield, Perceptron



Computational
Efficiency,
Mathematical
Tractability

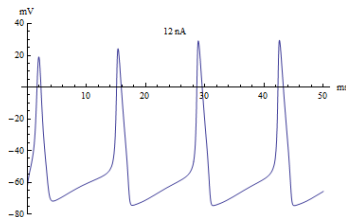
Phenomenological,
Qualitative

Cognitive Neuroscience, Psychology ...

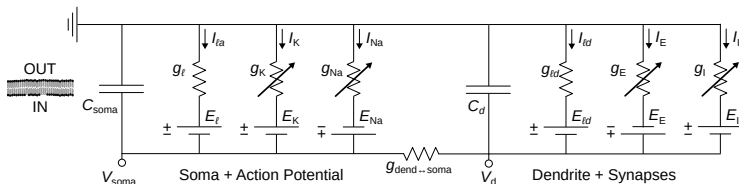
Conductance Model Point Neuron e.g. Hodgkin–Huxley

Nonlinear, action potential
costly to simulate

Study role of ion channels,
conductances, dendritic
morphology (shape), etc.



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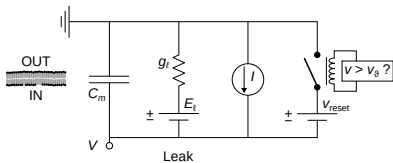
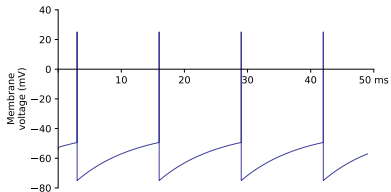


Week 3

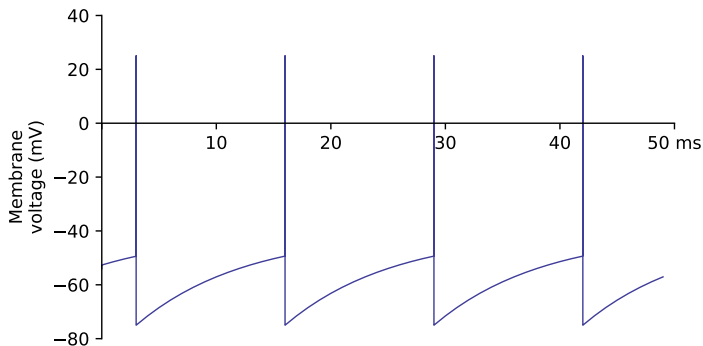
Leaky Integrate-and-Fire (LIF)

Departs from
physiology, but
sufficient to build
intuition

Easy to integrate



Leaky Integrate-and-Fire (LIF)



LIF model:

$$C\dot{v} = \frac{1}{R}(v_r - v) + I$$

if $v(t^-) > v_\theta$ then

Emit a spike

$$v(t^+) \leftarrow v_r$$

(if refractory) $v(t) = v_r$ for $t \in [t^+, t^+ + T_r]$

$v(t)$: membrane voltage

v_θ : Threshold (spike when $v(t) > v_\theta$)

v_r : Reset voltage ($v(t) \leftarrow v_r$ after spike)

R : Membrane resistance

C : Membrane capacitance

I : Current through the membrane

Simpler version (abstract) $\dot{v} = u - v$ $V_\theta = 1$ $V_r = 0$

Spikes if $v(t) \geq V_\theta$ for some future t .

Set $\dot{v} = 0$ & solve V_∞ . $0 = u - V_\infty \Rightarrow V_\infty = u \geq V_\theta = 1$
 $u \geq 1$ will spike.

If I start at $V_0 = V_r$, how long " T " to spike?

$$v(t) = V_0 + (1 - e^{-t/\tau})(V_\infty - V_0)$$

$$= (1 - e^{-t})u \quad \text{in this case}$$

When is $v(T) = V_\theta = 1$? $(1 - e^{-T})u = 1$

! the original version of this slide that you see in the lecture recording has a sign error that starts here. fixed

$$1 - e^{-T} = 1/u$$

$$e^{-T} = 1 - 1/u$$

$$-T = \ln\left(\frac{u-1}{u}\right)$$

$$T = \ln\left(u/(u-1)\right)$$